

Task 1: SSH Key-Based Authentication (RHEL 9.8)

OpenSSH Server is installed

- `dnf install openssh-server -y`

Start the SSH service

- `systemctl start sshd`

Enable SSH at boot

- `systemctl enable sshd`

Verify SSH service status

- `systemctl status sshd`

Test SSH to localhost

- `ssh localhost`

Type:

- `yes`

Task 2: Generate SSH Key Pair

Generate an RSA key pair:

- `ssh-keygen -t rsa -b 4096`

Press Enter for all prompts to accept the defaults.

Check:

- `ls -l ~/.ssh`

Add the Public Key

Append the public key to `authorized_keys`:

- `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`

Set the correct permissions:

- `chmod 700 ~/.ssh`
- `chmod 600 ~/.ssh/authorized_keys`

Verify:

- `ls -ld ~/.ssh`
- `ls -l ~/.ssh/authorized_keys`

Task 3: Disable Password Authentication

Open the SSH configuration file:

- `vi /etc/ssh/sshd_config`

Find or add the following lines:

- PasswordAuthentication no
- PubkeyAuthentication yes
- PermitRootLogin no

Save and exit (Esc, :wq).

Check the configuration:

- sshd -t

Restart SSH:

- systemctl restart sshd

Verify:

- systemctl status sshd

Test Key-Based Login

Connect to localhost:

- ssh localhost

To verify the authentication method:

- ssh -v localhost

Test Password Login (Should Fail)

Force password authentication:

- ssh -o PreferredAuthentications=password localhost